Chapter 9

Finite difference method in 1D

FDM 1D - goal

Goal: solve a 1D Poisson problem using finite differences

Poisson equation $\nabla^2\varphi = t$. **Hp**: Cartesian coordinates, $\partial\varphi/\partial y = \partial\varphi/\partial z = 0 \Rightarrow \varphi = \varphi(x)$:

$$\nabla^2\varphi = t \quad \rightarrow \quad \frac{\mathrm{d}^2\varphi}{\mathrm{d}x^2} = t, \quad x \in [a, b]$$

Problem (1): Dirichlet condition at a, Neumann condition at b

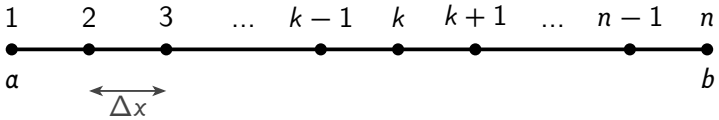
$$(1) \quad \begin{cases} \dfrac{\mathrm{d}^2\varphi}{\mathrm{d}x^2} = t & \forall x \in]a, b[\\ \varphi(a) = \varphi_a & x = a \quad \text{(Dirichlet)} \\ \left.\dfrac{\mathrm{d}\varphi}{\mathrm{d}n}\right|_b = \varphi'_b & x = b \quad \text{(Neumann)} \end{cases}$$

electrostatics: $t = -\rho/\varepsilon$; solution unique (Poisson problems chapter)

Goal: **discrete** formulation of (1): finite number of algebraic equations

FDM 1D - step 1: grid

Step 1: **grid** of nodes represents domain



n nodes with **uniform spacing**, $x_1 \equiv a$, $x_n \equiv b$:

$$\Delta x = \frac{L}{n-1}, \quad L = b - a$$

$n-1$: number of intervals

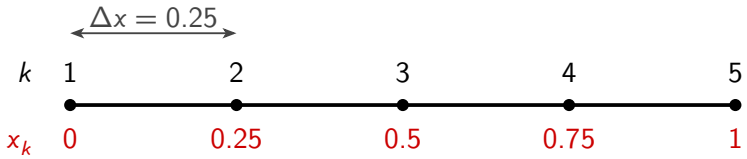
Linear mapping between coordinates and indices:

$$x_k = a + \Delta x(k-1)$$

e.g. $x_2 = a + \Delta x(2-1) = a + \Delta x$

FDM 1D - step 1: linear mapping

Example $[a, b] = [0, 1]$, $n = 5$ nodes



Spacing:

$$\Delta x = \frac{L}{n-1} = \frac{1}{5-1} = 0.25$$

Middle node, $k = 3$:

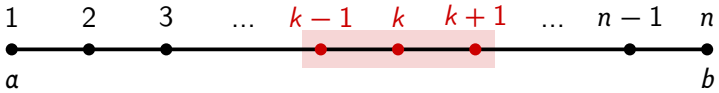
$$x_3 = a + \Delta x(k-1) = 0 + 0.25 \cdot (3-1) = 0.5$$

Last node, $k = n = 5$:

$$x_5 = 0 + 0.25 \cdot (5-1) = 1 = b$$

FDM 1D - step 2: internal nodes

Step 2: finite difference formulas express derivatives. Internal nodes $k = 2, \ldots, n-1$:



Equation at node k, centered formula for second derivative:

$$\left.\frac{\mathrm{d}^2\varphi}{\mathrm{d}x^2}\right|_k = t_k \quad \Rightarrow \quad \frac{\varphi_{k-1} - 2\varphi_k + \varphi_{k+1}}{\Delta x^2} + O(\Delta x^2) = t_k$$

Neglecting $O(\Delta x^2)$, values $\tilde{\varphi}_k$ satisfy:

$$\frac{\tilde{\varphi}_{k-1} - 2\tilde{\varphi}_k + \tilde{\varphi}_{k+1}}{\Delta x^2} = t_k$$

$\tilde{\varphi}_k$: approximation of unknown $\varphi(x_k)$, second-order accurate. Notation: φ_k instead of $\tilde{\varphi}_k$

FDM 1D - equations of internal nodes

Multiplying by Δx^2, one algebraic equation per internal node:

$$\begin{aligned}
\text{node } 2: &\quad \varphi_1 - 2\varphi_2 + \varphi_3 = t_2 \Delta x^2 \\
&\quad \vdots \\
\text{node } k: &\quad \varphi_{k-1} - 2\varphi_k + \varphi_{k+1} = t_k \Delta x^2 \\
&\quad \vdots \\
\text{node } n-1: &\quad \varphi_{n-2} - 2\varphi_{n-1} + \varphi_n = t_{n-1} \Delta x^2
\end{aligned}$$

$n-2$ equations, n unknowns $\varphi_1, \ldots, \varphi_n$

> Two missing equations: from **boundary conditions**

FDM 1D - boundary conditions

Node 1 ($x = a$), Dirichlet condition: value known

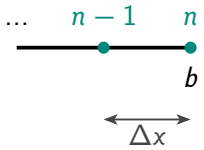
$$\varphi_1 = \varphi_a$$

Node n ($x = b$), Neumann condition. Condition to impose:

$$\left.\frac{\mathrm{d}\varphi}{\mathrm{d}x}\right|_b = \varphi'_b \quad (\varphi'_b \text{ known})$$

Unknowns: nodal values φ_k, not derivatives $\Rightarrow$ express $\left.\frac{\mathrm{d}\varphi}{\mathrm{d}x}\right|_b$ through φ_k with finite difference formula. Only nodes up to n exist: backward formula

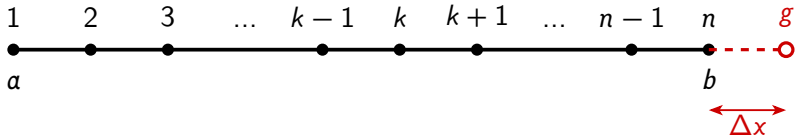
$$\left.\frac{\mathrm{d}\varphi}{\mathrm{d}x}\right|_b = \frac{\varphi_n - \varphi_{n-1}}{\Delta x} + O(\Delta x) = \varphi'_b$$



First-order accurate, while internal equations are second-order

FDM 1D - ghost node technique

Question: how to use centered formula for $\mathrm{d}\varphi/\mathrm{d}x$ at b?



Add fictitious **ghost node** g at $x_g = b + \Delta x$, outside domain

Two equations at node n, centered formulas:

$$\begin{cases} \left.\frac{\mathrm{d}\varphi}{\mathrm{d}x}\right|_b = \dfrac{\varphi_g - \varphi_{n-1}}{2\Delta x} = \varphi'_b & [1] \quad \text{Neumann condition} \\ \varphi_{n-1} - 2\varphi_n + \varphi_g = t_n \Delta x^2 & [2] \quad \text{equation at node } n \text{ , } \left.\dfrac{\mathrm{d}^2\varphi}{\mathrm{d}x^2}\right|_n = t_n \end{cases}$$

both second-order accurate; one extra unknown φ_g, one extra equation

FDM 1D - ghost node elimination

From [1] (Neumann condition), deriving φ_g:

$$\varphi_g - \varphi_{n-1} = 2\Delta x \varphi'_b \quad \Rightarrow \quad \varphi_g = \varphi_{n-1} + 2\Delta x \varphi'_b$$

Substituting into [2] (equation at node n):

$$\varphi_{n-1} - 2\varphi_n + \varphi_{n-1} + 2\Delta x \varphi'_b = t_n \Delta x^2$$

Isolating unknowns φ_{n-1}, φ_n:

$$\varphi_{n-1} - \varphi_n = \Delta x \left(\tfrac{\Delta x}{2} t_n - \varphi'_b \right)$$

algebraic equation for node n, second-order accurate

From previous slide

$$\begin{cases} \dfrac{\varphi_g - \varphi_{n-1}}{2\Delta x} = \varphi'_b & [1] \\ \varphi_{n-1} - 2\varphi_n + \varphi_g = t_n \Delta x^2 & [2] \end{cases}$$

[1] Neumann condition
[2] equation at node n

FDM 1D - linear system

Assembling algebraic equations: discrete version of (1)

node 1: $\varphi_1 = \varphi_a$

nodes $2, \ldots, n-1$: $\varphi_{k-1} - 2\varphi_k + \varphi_{k+1} = t_k \Delta x^2$

node n: $\varphi_{n-1} - \varphi_n = \Delta x(\Delta x/2\ t_n - \varphi_b')$

$$\underbrace{\begin{bmatrix} 1 & 0 & 0 & \ldots & 0 \\ 1 & -2 & 1 & \ldots & 0 \\ \vdots & \ddots & \ddots & \ddots & \vdots \\ 0 & \ldots & 1 & -2 & 1 \\ 0 & \ldots & 0 & 1 & -1 \end{bmatrix}}_{[K]} \underbrace{\begin{Bmatrix} \varphi_1 \\ \varphi_2 \\ \vdots \\ \varphi_{n-1} \\ \varphi_n \end{Bmatrix}}_{\{\varphi\}} = \underbrace{\begin{Bmatrix} \varphi_a \\ t_2 \Delta x^2 \\ \vdots \\ t_{n-1} \Delta x^2 \\ \Delta x(\Delta x/2\ t_n - \varphi_b') \end{Bmatrix}}_{\{\text{Rhs}\}}$$

FDM 1D - sparse matrices

$[K]$ is **tri-diagonal**: non-zero entries only on main diagonal and two adjacent ones

$[K]$ is **sparse**: most entries zero

Example $n = 1000$: $[K]$ has 10^6 entries, but at most 3 non-zeros per row

$\rightarrow$ about $3 \cdot 10^3$ non-zeros, i.e. $\dfrac{3 \cdot 10^3}{10^6} \cdot 100 = 0.3\%$ of the entries

> Property of FDM: each equation involves only neighbouring nodes
>
> $\Rightarrow$ FDM produces **sparse** matrices

Only non-zeros stored; dedicated solvers exploit structure

tri-diagonal system: $O(n)$ operations, instead of $O(n^3)$ for full matrix

Diagonally dominant matrices - definitions

Consider a square matrix $[A]$ with n rows, entries $a_{i,j}$

Row i is **strictly** / **weakly** dominant if:

$$|a_{i,i}| > \sum_{j=1, j\neq i}^{n} |a_{i,j}| \quad \text{strictly,} \qquad |a_{i,i}| \geq \sum_{j=1, j\neq i}^{n} |a_{i,j}| \quad \text{weakly}$$

$a_{i,i}$: entry on main diagonal; $a_{i,j}$, $j \neq i$: other entries of row i

$$\begin{bmatrix} a_{1,1} & a_{1,2} & a_{1,3} \\ a_{2,1} & a_{2,2} & a_{2,3} \\ a_{3,1} & a_{3,2} & a_{3,3} \end{bmatrix}$$

row $i = 2$

Matrix $[A]$ diagonally dominant:

- **strictly**: **every** row strictly dominant
- **weakly**: every row **at least** weakly dominant (some may be strict)

e.g. all rows strict but one weak $\Rightarrow$ weakly dominant

e.g. row $2 = (1, -5, 2)$:

$$|-5| > |1| + |2| = 3$$

row 2 strictly dominant

Diagonally dominant matrices - example

Example $[K]$ of 1D problem with $n = 4$

$$[K] = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 1 & -2 & 1 & 0 \\ 0 & 1 & -2 & 1 \\ 0 & 0 & 1 & -1 \end{bmatrix}$$

row 1: $|1| > 0$ strict

rows 2, 3: $|-2| = 1 + 1$

row 4: $|-1| = 1$

> $[K]$ is **not** strictly diagonally dominant, but **weakly** diagonally dominant

Diagonally dominant matrices - unique solution

Question: when does $[A]\{x\} = \{b\}$ have a **unique** solution?

Theorem 1 $[A]$ strictly diagonally dominant $\Rightarrow$ unique solution

Theorem 2 $[A]$ weakly diagonally dominant, **at least one** strictly dominant row, reachable from **every** row $\Rightarrow$ unique solution

reachable: $a_{i,j} \neq 0$ lets us step from row i to row j (equation i contains unknown x_j);
a chain of such steps leads from every row to a strictly dominant one

Theorem 1: Gershgorin's circle theorem; Theorem 2: weakly chained diagonally dominant matrices

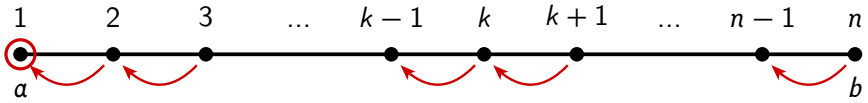
FDM 1D - unique discrete solution

Apply Theorem 2 to $[K]$

$[K]$ weakly dominant; row 1 (**Dirichlet**) strictly dominant

Row 1 reachable from every row:
equation k contains φ_{k-1}
$(a_{k,k-1} = 1 \neq 0)$
$\Rightarrow$ step $k \to k-1 \to \ldots \to 1$

node n:	$\varphi_{n-1} - \varphi_n = \Delta x(\Delta x/2\ t_n - \varphi'_b)$	$\to n-1$
node k:	$\varphi_{k-1} - 2\varphi_k + \varphi_{k+1} = t_k \Delta x^2$	$\to k-1$
node 2:	$\varphi_1 - 2\varphi_2 + \varphi_3 = t_2 \Delta x^2$	$\to 1$
node 1:	$\varphi_1 = \varphi_a$	strictly dominant



One **Dirichlet** node $\Rightarrow$ **unique** solution

FDM 1D - Neumann conditions at both ends

Assume Neumann conditions at a and b, both with ghost node

At a: same steps as at b, ghost node at $a - \Delta x$ (right)

$$\varphi'_a = \left.\frac{\mathrm{d}\varphi}{\mathrm{d}n}\right|_a = -\left.\frac{\mathrm{d}\varphi}{\mathrm{d}x}\right|_a, \text{ outward normal along } -x$$

No strictly dominant row. Every row of $[K]$ sums to zero:

$$[K]\begin{Bmatrix}1\\1\\\vdots\\1\end{Bmatrix} = \begin{Bmatrix}-1+1\\1-2+1\\\vdots\\1-1\end{Bmatrix} = \{0\}$$

Node 1 (ghost node elimination)

$$\begin{cases} -\dfrac{\varphi_2 - \varphi_g}{2\Delta x} = \varphi'_a & [1] \\ \varphi_g - 2\varphi_1 + \varphi_2 = t_1 \Delta x^2 & [2] \end{cases}$$

φ_g from [1], substituted into [2]:

$$-\varphi_1 + \varphi_2 = \Delta x \left(\frac{\Delta x}{2} t_1 - \varphi'_a \right)$$

$[K]$ **singular**: if $\{\varphi\}$ is a solution, $\{\varphi\} + c\{1\}$ is a solution too

Same as continuous problem: Neumann conditions only $\Rightarrow$ φ unique **up to a constant**

FDM 1D - Robin condition

Assume Robin condition at b: $\varphi + \beta \frac{\mathrm{d}\varphi}{\mathrm{d}n}\Big|_b = 0$ (conduction chapter: $\beta = R\sigma A_R > 0$)

From [1] with $\varphi'_b = -\varphi_n/\beta$:

$$\varphi_g = \varphi_{n-1} - 2\Delta x \varphi_n/\beta$$

Substituting into [2] and dividing by 2:

$$\varphi_{n-1} - \left(1 + \frac{\Delta x}{\beta}\right)\varphi_n = \frac{\Delta x^2}{2} t_n$$

Node n (ghost node elimination)

$$\begin{cases} \dfrac{\varphi_g - \varphi_{n-1}}{2\Delta x} = \varphi'_b & [1] \\ \varphi_{n-1} - 2\varphi_n + \varphi_g = t_n \Delta x^2 & [2] \end{cases}$$

φ_g from [1], substituted into [2]:

$$\varphi_{n-1} - \varphi_n = \Delta x \left(\frac{\Delta x}{2} t_n - \varphi'_b\right)$$

$\beta > 0$: $|1 + \Delta x/\beta| > 1$, row n **strictly** dominant
$\beta < 0$: $|1 + \Delta x/\beta| < 1$ for small Δx, dominance is lost

Same condition as continuous uniqueness theorem: $\alpha/\beta > 0$

FDM 1D - post-processing: gradient

Goal: from nodal values φ_k, compute $\nabla\varphi$ (e.g. $\boldsymbol{E} = -\nabla\varphi$). In 1D: $\nabla\varphi \to \mathrm{d}\varphi/\,\mathrm{d}x$

Internal nodes $k = 2, \ldots, n-1$: centered formula

$$\left.\frac{\mathrm{d}\varphi}{\mathrm{d}x}\right|_k = \frac{\varphi_{k+1} - \varphi_{k-1}}{2\Delta x} + O(\Delta x^2)$$

Boundary nodes: only one neighbour, **one-sided** formulas

$$\text{node } 1: \quad \left.\frac{\mathrm{d}\varphi}{\mathrm{d}x}\right|_1 = \frac{\varphi_2 - \varphi_1}{\Delta x} + O(\Delta x), \qquad \text{node } n: \quad \left.\frac{\mathrm{d}\varphi}{\mathrm{d}x}\right|_n = \frac{\varphi_n - \varphi_{n-1}}{\Delta x} + O(\Delta x)$$

forward formula at node 1, backward formula at node n

Homework: rewrite linear system imposing Neumann condition at b with **second-order one-sided** formula (no ghost node). Is $[K]$ still tri-diagonal?

one-sided formulas: Numerical differentiation chapter

Lab - FDM for the 1D Poisson equation (in class)

Done in class, in MATLAB

Folder: `03_fdm_1D`

Goal: Dirichlet and Neumann conditions, post-processing, convergence test

FDM 1D - summary

- Grid of n nodes; unknowns: the nodal values φ_k
- Internal nodes: centered formula, $O(\Delta x^2)$; Dirichlet: known value; Neumann: ghost node to keep $O(\Delta x^2)$
- Sparse (tri-diagonal) linear system $[K]\{\varphi\} = \{\text{Rhs}\}$
- Uniqueness of the discrete solution mirrors the continuous one: Dirichlet $\rightarrow$ unique; Neumann only $\rightarrow$ up to a constant
- Verification: test problems with a known solution, and the error ratio when halving Δx